# Towards Melodic Extension Using Genetic Algorithms

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**Abstract**

In this paper we will discuss the melodic features used as the basis for a Genetic Algorithm fitness function and mutation procedures. We will discuss the method of selecting the features and what they reveal about melodic writing and genetic musical processes. Genetic Algorithms are considered promising for music composition because they combine 'creativity' (ability to explore a large search space) with constraints (a fitness function 'weeds out' creative 'excess'). However, progress to date has been limited primarily because it is difficult to define fitness functions which capture the aesthetic qualities of the wide gamut of successful melodies. In this paper we report on research that addresses this problem for a modest compositional task, melodic extension. We will discuss how features were chosen, measured for significance, and might be combined to form the basis for a fitness function. We discuss how the features are implemented in knowledge-enhanced mutation and crossover operators.

**Melodic extension**

The use of heuristic principles for automated music composition is well established in computer music (Moorer 1972, Laske 1992) but the use of Genetic Algorithms in this task is less well established. They have been used previously in the harmonisation of melodies by McIntyre (1994), Horner and Ayers (1995), and Wiggens et al. (1997), an interactive system was developed by Biles (1994) for performance of jazz improvisations, and have been used for automated composition by Horner and Goldberg (1991) and Jacob (1995), whose systems attempted autonomous composition, rather than computer assisted composition.

This paper reports on analysis research that is part of an investigation into how secondary music students can be assisted in learning to compose melodies by working with a computer system that actively suggests extensions to, and can provide critiques of, a student's composed melody. The complete investigation will explore the suitability of Genetic Algorithms to the 'fuzzy' task of generating 'appropriate' melodic extensions. These melodic suggestions will provide 'timely' interactive support to the students as rated by the Genetic Algorithm fitness function. The first step in achieving an effective fitness function is the definition of the features of a 'good' melody. This paper outlines the process and results of that melodic feature analysis.

The recent work of Cope (1997) with his CUE software is the research most closely related to this project in that Cope's goal was similar—to provide generative extensions to partially composed material in common music notation. Our project differs from Cope's work in its use of Genetic Algorithms rather than Associative Transition Networks (ATN) for melodic generation. We feel that this approach will provide a system at least as customable as CUE with stable performance efficiency, rather than the problems of scalability inherent in increasing database size using ATNs.

Because the analysis study presented here is the first part of a project creating music which extends monophonic musical material, it is harmonically static and does not aim to accompany any predefined harmonisation, rather, the referent for melodic continuance is the temporal context where similarity and symmetry of features (continuance) is valued. The analysis project aimed to extrapolate features from existing musical material in an attempt to provide musically appropriate features which capture a balance of novelty and consistency.

The analysis is not intended to be an exhaustive study of melody writing but rather limited to the world of short melodies in the tradition of Western diatonic music. In keeping with this aim, some of the assumed restrictions in this study include that; the compositional activity is monophonic, the representational system is common practice notation, the style of music is assumed to be western, diatonic, rhythmically regular using only simple time signatures and no modulation of key, and the melodic suggestions need to be 'appropriate' not ideal. The features should reinforce the music theory 'rules' in the school curriculum, rather than being overly 'novel' or 'creative'.

The software environment for this project is the Java-based jMusic composition language which is an ongoing research project of the QUT music department. This language provides a music data structure and visualisation interface for notation and MIDI file reading and writing routines for accessing the sample melodies for analysis.

**Possible melodic features**

In order to have the analysis find features which will be reinforced in school music programs, a variety of educational publications on melody writing were reviewed when collecting the melodic features. From these sources the following list of features were defined as likely to indicate tendencies in melody. The melodic features were divided into several broad categories; counts (the number of discrete elements, such as pitches), proportion (the degree to which elements are represented), ranges (the diversity of elements), pitch movement (melodic stability and contour), rhythmic stability (degree of syncopation), and patterns (the amount of internal repetition or consistency).

The complete list of melodic features and their definitions are listed below. Each feature is represented as a ratio (x per y, or x over y) and the analysis output results in a real number between 0.0 and 1.0 for each feature of each melody.

Counts
======

(1) Note Density
The number of notes per quanta (Quantum is a unit representing the smallest allowed Rhythm Value—semiquaver beats). This feature indicates the degree of business and sparseness of a melody.

(2) Pitch Variety
The number of distinct pitches per note. This feature indicates the diversity of the pitch class set used in writing the melody.

(3) Rhythmic Variety
The distinct rhythm values per value. It measures the extent to which the 16 rhythmic values between semiquaver and semibreve are used. This feature gives some indication of the rhythmic coherence of the piece and the degree to which it is unsettled or varied.

(4) Climax Strength
How often the highest pitch is used. This value is divided by one such that reaching ht high point once score high and more often lessens its impact.

Proportion of total rhythmic time
==========

(5) Rest Density

The number of rests per quanta (semiquaver beats). This feature indicates the degree of silence or sparseness of a melody. It is related to Note Density, (1) above.

(6) Tonal Deviation
The number of non-scale (ie., major or harmonic minor) pitches per quanta. This feature provides an indication of how strongly tonal the melody is.

(7) Key Centeredness
The number of primary note quanta per quanta. This feature checks the number of semiquaver beats allocated to the root and dominant notes. It provides an indication of how strongly the melody has a sense of key.

Ranges
======
(8) Pitch Range
The pitch span divided by the possible pitch span. This feature provides an indication of the extent of the melodic contour. For our purposes the maximum range was set to two octaves, which may seem narrow but greater values resulted in too narrow a standard deviation to be useful.

(9) Rhythmic Range
The rhythmic value range over the maximum possible rhythmic value range. This measures the extent to which the melody utilises a wide variance of rhythmic values. It is different from the rhythmic variety in that it takes account of the quantity of the different rhythm values, not just their occurrence. There were sixteen allowable rhythm values from semibreve to semiquaver.

Pitch Movement
==========
(10) Repeated pitch
This feature counts the consecutive notes of same pitch. It measures the lack of movement in the melody. Note that for this feature rests are ignored ie., it considers all note to next note pairs.
(11) Repeated rhythmic value
This feature counts consecutive notes of same duration. It measures the rhythmic consistency in the melody. Note that for this feature rests are ignored ie., it considers all note to next note pairs.

(12) Melodic Direction Stability
The pitch steps in the same direction compared to the number of steps. This checks the degree of melodic contour variation. The feature measure pitch intervals, which means that three notes (two intervals) are required to measure a pattern. Rests are ignored in this feature.

(13) Overall Pitch Direction
The sum of rising intervals over the sum of all intervals. This feature checks the overall tendency of the melodic contour. A melody starting and finishing on the same note will score 0.5, higher indicates an overall rise, and lower scores indicate a descending contour. Rests are ignored in this feature.

(14) Movement by Step
The number of pitch movements by diatonic step over the total number of steps. This indicates the smoothness of the melody. A high score indicates a smooth melodic curve with few large leaps. A step interval is any diatonic interval and might be one or two semitones. Rests are ignored in this feature.

(15) Dissonant Intervals
The number of dissonant intervals over the total number of intervals. The dissonance rating of an interval for this feature can in be one of three classes:

| | |
|---|---|
| 0, 5, 7 | 0.0 (includes unison, $4^{th}$ and $5^{th}$) |
| 2, 3, 4, 8, 9, 10 | 0.5 (includes Maj $2^{nd}$, $3^{rds}$, $6^{ths}$, and min $7^{th}$) |
| 1, 6, 11 | 1.0 (includes semitone, tritone, Maj $7^{th}$) |

Rests are ignored in this feature.

(16) Leap Return Rule
The number of pitch interval direction reversals after large leaps over the total number of large leaps. A large leap is greater than or equal to eight semitones (minor $6^{th}$).

The returning interval must be less than the jump interval preceding it.

Syncopation
==========

(17) Syncopation
The beat quanta on which notes start compared with the total number of beats. This feature indicates the degree to which the beat pulse of the music is maintained. Rests are differentiated from notes in this feature.

Patterns
======

(18) Repeated Pitch Patterns
The number of pitch patterns which are repeated over the number of quanta. This feature compares intervallic relationships so that sequences count as repeats, but at present sequences are absolute rather than diatonic. Also, to cut computation, we only consider sequences of 3 and 4 intervals starting on the beat, but attempt to find a match at every quantum. Rests are taken into account such that they may be part of a pattern.

(19) Repeated Rhythm Patterns
The number of rhythm patterns with repetitions over the number of quanta. This feature measures the rhythmic coherence of the melody as developed through repetition. The rhythmic ratio is checked rather than absolute rhythms, therefore, a pattern crotchet, minim, crotchet will match quaver, crotchet, quaver. To cut computation, we only consider sequences of 3 and 4 intervals starting on the beat, but we attempt to find a match at every quantum so that offset or phased repetitions are not ignored. Rests are taken into account such that they may be part of a pattern.

A list of definitions for terms can be found at the end of the paper.

**Analysis procedure**

The analysis of features used MIDI files as a readily available source of melodic data. Code was written in the Java language, using the jMusic MIDI reading and writing libraries, to handle the analysis process. The interface for the analysis program is shown in figure 1. Melodies from the Western art music repertoire were sourced as MIDI files and were appropriately prepared for analysis by quantising rhythms, transposing into C major or A minor, and setting playback to 120 beats per minute. This normalised the data such that reasonable comparisons could be drawn between them.

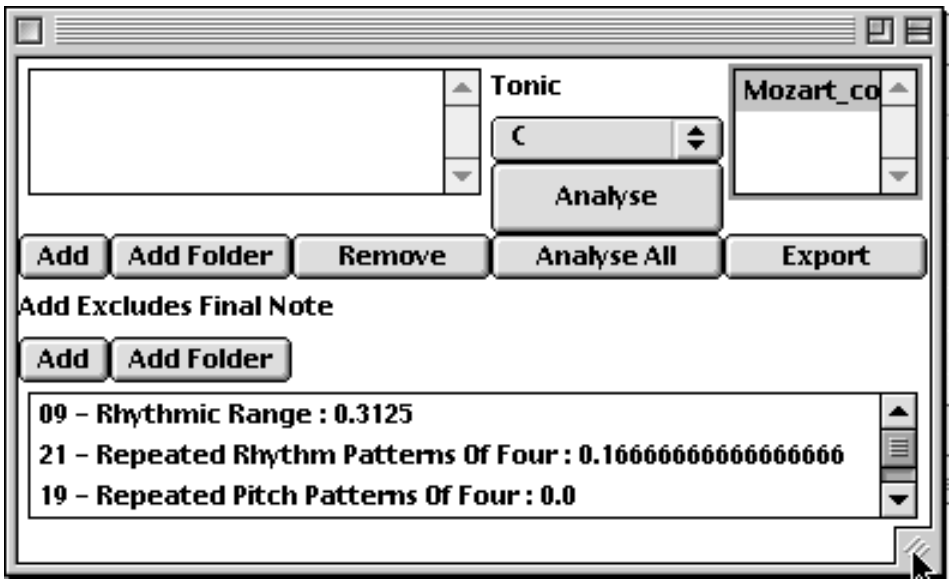



Fig. 1 – *Melodic Analysis Application Interface*

The repertoire list comprised melodies by the following composers;

- Bach
- Beethoven
- Du Fay
- Gesualdo
- Gibbons
- Montiverdi
- Mozart
- Palestrina
- Tchaikovsky
- Traditional Nursery Rhymes

Features were calculated on each melody and the average resalt and standard deviation of each feature was calculated for the whole group and various subgroups, as described below. In order to determine which features might be most important in identifying well formed melodies, T tests were run and significance assessed.

**Results**

We analysed a total of 31 melodies categorised according to style and mode. There were 17 melodies from classical compositions, 9 from early music compositions and 5 nursery rhymes. Eight of the melodies were in a minor key and the remainder were in major key. There were no significant differences in feature values between the major and minor categories. These categories were chosen because, to the researchers' ear, distinctive differences in melodies appear along these stylistic lines. In particular, there are differences from the harmonic and rhythmic simplicity of nursery rhymes and the modal 'flavour' of Early music. Another consideration was that the teaching of melody writing emphasises 'rules' derived from repertoire in these styles. There were some statistically significant differences between the three categories of melodic style.

Early music melodies had a significantly lower note density (Feature 1, t(df=24) = 2.89, significant @ 0.005) but a significantly higher rhythmic variety. This later characteristic was evidenced by a high value for rhythmic variety (Feature 3, t(df=12) = 2.12, significant @ 0.05), a high value for rhythmic range (Feature 9, t(df=12) = 1.93, significant @ 0.05) and the lowest rate of repeated note duration (Feature 11). Perhaps paradoxically, this greater rhythmic variety is due to more frequent use of long notes such as semibreves and breves in early music. By contrast, Classical melodies had the highest rate of repeated note durations.

There were no significant differences between the three categories of melody in their Pitch Variety (Feature 2) or in their Key Centredness (Feature 7). Tonic and dominant pitches (the pitches chosen as determining key centredness) constituted some 40% of quanta in all three categories.

Nursery rhymes used only scale notes in their melodies (Feature 6) and on average, repeated their climatic note twice (Feature 4). By comparison, the other two categories used non-scale notes for about 5% of quanta and the climatic note was repeated with less frequency. However, the rate of consecutive repeated notes (ie., having same pitch, Feature 10) was significantly lower in nursery rhymes than the other categories (t(df = 12) = 1.45, significance < 10%).

Pitch Range (Feature 8) was close to an octave for all three categories, slightly exceeding the octave for classical compositions and slightly under for nursery rhymes. Similarly, there were no significant differences between categories for stability of melodic direction (Feature 12), indicating that most melodies are generally symmetrical in their use of ascending and descending leaps. Although overall pitch direction (Feature 13) has an average value close to 0.5 for all categories, nevertheless, nursery rhymes have a significantly greater tendency to fall sightly than do Early Music melodies. (t(df=12) = 2.64, significance @5%).

With regard to Movement by Step (Feature 14), 93% of the intervals in nursery rhymes move by step which is significantly greater than the value of 75% for classical melodies and 78% for early music melodies. Nursery rhymes also have twice the rate of repeated 4 note pitch patterns (Feature 18) than the other two categories, (a significant difference) while classical melodies have the highest rate of repeated rhythmic patterns (Feature 19) although the differences for this feature are not significant due to high variability within melodic styles.

**Towards a GA fitness function**

The (normalised) quantitative melodic features we have described above were developed based on rules of best practice in melodic writing as taught to students of melody. While these features are 'obvious' in their application of established melody writing curricular, this analysis points toward the statistical integrity of the features and thus how reliable they will be in an computer-based judgement of melodic 'fitness'. Our use of the features is intended to modulate or constrain the application of a fitness function in a Genetic Algorithm. To be of practical use in this context, it is probably useful to have a variety of quantitative features that differ in their range of acceptable values, that is, which differ in their standard deviation. The lower the standard deviation of a feature the more severe the constraint it places on an acceptable melody. For example Feature 13, Overall pitch direction, has a comparatively low standard deviation and will place strong constraint on the selection and mutation of melodies. By contrast, Feature 19 has a large standard deviation, and may be useful only to remove extremely "dysfunctional" melodies from a population. The combination of two such features helps the Genetic Algorithm to gradually evolve a population of "fit" melodies.

From these results, it appears that the following functions will be useful in constructing a Genetic Algorithm fitness function;
(2) Pitch Variety
(4) Climax Strength
(6) Tonal Deviation
(7) Key Centeredness
(8) Pitch Range
(12) Melodic Direction Stability

This next group of functions seems less significant but somewhat indicative, and might contribute to a fitness function with small weightings.
(10) Repeated pitch
(14) Movement by Step
(18) Repeated Pitch Patterns
(19) Repeated Rhythm Patterns

The possibilities of weighting and coordination of features into a fitness function are numerous and the topic of future extensions to this research project.

**Summary**

In this paper we have reported on the analysis of melodic features as a determinant for their application in a Genetic Algorithm fitness function designed to assess computer generated melodic extensions. We have specified a series of melodic features, based upon melody writing curriculum used in schools, and defined those features as normalised quantitative measures, as required for computer implementation. The process of analysing the features against a set of repertoire in western musical styles was described and the significant results reported. The features which appear useful for inclusion in a fitness function have been identified.

Further research is required to determine the effective weighting of the features and any significant co-occurrence of features which might be useful. A number of features in this process produced ambiguous results (including Features 1, 9, 7, and 13) and some changes to the analysis implementation for those features and the

testing a different melody repertoire may clarify the usefulness of those features.
It is clear that although the human application of melody writing 'rules' seems straight forward, the articulation of those rules for autonomous computer assessment, as required for a Genetic Algorithm fitness function, is less obvious. Through this research we have described those rules as features, and identified the feature descriptions which are statistically reliable against a set of melodies from respected composers. We look forward to continuing this research and reporting on how effective the features descriptions are in selecting respectable melodies of algorithmic than genetic origin.

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**References**

Biles, J. A. 1994. GenJam: A genetic algorithm for generating jazz solos. In ICMC Proceedings 1994. The International Computer Music Association.

Cope, D. 1997. "The Composers Underscoring Environment: CUE" in *Computer Music Journal*, Vol. 21, No. 3, Fall 1997, pp. 20-37

Horner, A. & Ayers, L. 1995. Harmonisation of musical progression with genetic algorithms. In ICMC Proceedings 1995. pp. 483-484. The International Computer Music Association.

JMusic. 2000. Sorensen, A and Brown, A. http://www.academy.qut.edu.au/music/newmedia/jmusic/

Laske, O. E. 1992. Artificial Intelligence and Music: A cornerstone of cognitive musicology. In, Balban, M., Ebcioglu, K. & Laske, O. Understanding Music with AI: Perspectives on music cognition. pp. 3-29. Menlo Park, CA: The AAAI Press.

Moorer, J. A. 1972. Music and Computer Composition. Reprinted in Schwanauer, S. & Levitt, D. 1993. Machine Models of Music. pp. 167-188. Cambridge, Massachusetts: The MIT Press.

McIntyre, R. A. 1995. Composition with genetic algorithms. Technical report, University of Michigan.